Academic Year 2025/2026 (2nd Term)
ARTI-308: Machine learning

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Oil and Gas Pipeline Predictive Maintenance

Machine learning

ARTI 308 – 6A01 – Group 1

Submitted to: College of Computer Science and Information Technology.

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Abstract

The oil and gas industry relies on pipeline networks that face constant threats from high pressure and corrosion. This project proposes a Machine Learning model to shift from inefficient traditional maintenance to a data-driven, proactive approach. We developed a multi-class classification system to predict if pipeline segments are in Normal, Moderate, or Critical states. We developed this model to avoid unexpected failures that cause catastrophic environmental damage and financial loss, which current reactive methods fail to prevent efficiently. We envisage an outcome that reduces maintenance costs by 20-30% while significantly enhancing operational safety and reliability.

Introduction

The oil and gas industry depends heavily on pipeline systems for transporting petroleum products across long distances. These pipelines operate under harsh environmental and operational conditions such as high pressure, corrosion, temperature fluctuations, and material degradation. Over time, these factors can weaken pipeline structures and increase the risk of leakage or failure, leading to environmental damage, safety hazards, operational downtime, and significant financial losses. Therefore, maintaining pipeline integrity and reliability has become one of the major challenges in the energy sector. Traditional maintenance approaches mainly rely on reactive maintenance or fixed schedule-based maintenance, which are often inefficient because failures may occur unexpectedly before maintenance actions are performed. With the increasing availability of industrial operational data, Machine Learning (ML) techniques have emerged as intelligent solutions for predictive maintenance by identifying degradation patterns and predicting failures before they occur [4], [8].

Several previous studies applied machine learning techniques to corrosion prediction and pipeline monitoring. Fu et al. [1] proposed an SSA-LightGBM model for corrosion-rate prediction, while Wang et al. [2] developed a hybrid deep learning framework combining Random Forest, BiLSTM, and GRU models. Xie et al. [3] introduced a knowledge graph and neural network approach to improve corrosion prediction accuracy. Although these studies achieved promising results, most focused on corrosion-rate regression rather than direct maintenance decision classification suitable for practical industrial applications. To address this gap, this study proposes a machine learning-based predictive maintenance framework for oil and gas pipelines using a publicly available Kaggle dataset [5]. The objective of this project is to classify pipeline conditions into three maintenance categories: Normal, Moderate, and Critical using operational and degradation-related features such as pipe size, wall thickness, pressure, temperature, corrosion impact percentage, material loss, and pipeline age.

This study investigates two supervised machine learning classification techniques: Random Forest (RF) and Support Vector Machine (SVM). Random Forest was selected because of its ability to handle noisy and imbalanced datasets while reducing overfitting through ensemble learning [7], [10], while SVM was selected because of its strong classification capability and effectiveness in high-dimensional feature spaces [6]. Several preprocessing and feature engineering techniques were applied, including label encoding, feature scaling, removal of leakage-related features, and creation of engineered features to improve predictive performance. The dataset was divided using an 80/20 stratified train-test split, and Stratified K-Fold Cross-Validation was applied to improve model reliability. The models were evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Experimental results showed that the Random Forest model achieved the best performance with a testing accuracy of 92.5%, outperforming the SVM model which achieved 82.5% accuracy. The remainder of this paper is organized as follows. Section 2 presents the related literature review, Section 3 describes the proposed techniques, Section 4 explains the experimental setup, Section 5 presents the results and discussion, and Section 6 concludes the study and provides recommendations for future work.

2. Review of Related Literatures

Machine learning has emerged as a key enabling technology in predictive maintenance applications for oil and gas pipelines, particularly due to its capability to model nonlinear degradation behaviors and support early failure detection. Recent research has primarily focused on corrosion-rate prediction, pipeline integrity assessment, and hybrid optimization-based modeling techniques, while relatively limited attention has been given to direct maintenance decision classification tasks.

Fu et al. [1] proposed an SSA-LightGBM model for predicting the average corrosion rate in buried oil and gas pipelines. Their approach demonstrated strong predictive performance through hyperparameter optimization, highlighting the effectiveness of gradient boosting techniques in corrosion modeling. Similarly, Wang et al. [2] introduced a hybrid deep learning framework combining Random Forest, IBWO optimization, BiLSTM, and GRU networks, achieving $R^2 > 0.99$, which confirms the ability of ensemble deep architectures to capture complex corrosion patterns. In another study, Xie et al. [3] developed a knowledge-graph-enhanced neural network optimized using PSO and GA algorithms, showing that integrating domain knowledge can significantly improve corrosion prediction accuracy. Although these studies achieve high performance, they are primarily limited to regression-based corrosion rate prediction rather than operational maintenance decision-making.

At a broader level, Nwulu et al. [4] reviewed machine learning applications in predictive maintenance within the oil and gas sector. The study highlights the effectiveness of models such as decision trees, random forests, neural networks, and ensemble methods in

improving maintenance efficiency and reducing downtime. However, it also emphasizes the lack of standardized and publicly available datasets, which limits reproducibility and benchmarking across studies.

The dataset used in this study—Predictive Maintenance for Oil & Gas Pipelines (Kaggle) [5]—contains key features such as corrosion impact, material loss, thickness reduction, and a binary target variable (maintenance_required). Unlike most datasets used in previous studies, this dataset enables a direct classification formulation for maintenance decision prediction rather than continuous corrosion estimation. To the best of the authors' knowledge based on searches in Google Scholar and available academic databases, no peer-reviewed studies have explicitly reported the use of this Kaggle dataset for maintenance classification tasks.

Although existing literature focuses extensively on corrosion-rate prediction using proprietary datasets, many of the extracted features (e.g., corrosion severity, material degradation, and environmental indicators) are conceptually aligned with the variables present in the Kaggle dataset used in this study. However, there remains a clear gap in translating these predictive models into binary maintenance decision systems suitable for operational use.

Study	Data	Methods	Task	Findings	Identified Gap
[1] Fu et al., 2025	Buried pipeline corrosion data	SSA-LightGBM	Corrosion-rate regression	High accuracy with optimized boosting	No maintenance decision classification
[2] Wang et al., 2024	Pipeline corrosion dataset	RF + IBWO + BiLSTM + GRU	Corrosion-rate regression	R^2 > 0.99 strong predictive performance	Regression only; no operational decision output
[3] Xie et al., 2024	Corrosion + knowledge graph data	KG + Neural Network (PSO, GA)	Corrosion-rate regression	Improved accuracy via domain knowledge	No binary maintenance prediction
[4] Nwulu et al., 2023	Oil & gas PdM literature	Review study	Conceptual overview	ML improves maintenance efficiency	No experimental validation

Table 1: Summary of related work and identified gaps

Despite the increasing number of studies in this field, no peer-reviewed research has been found that uses the exact Predictive Maintenance for Oil & Gas Pipelines (Kaggle)dataset adopted in this study, as most prior work relies on different or proprietary datasets. However, one informal implementation (on Kaggle) was identified that applied machine learning models to this dataset without a formal publication, and the following table presents a comparison between that work and our study.

Aspect	Kaggle Notebook	This Study
Dataset	**Same Kaggle dataset**	**Same Kaggle dataset**
Feature Handling	**All features kept (possible leakage)**	**Removed leakage-prone features**
Models Used	**RF, XGBoost, GBM, Logistic Regression**	**Random Forest, SVM**
Best Model	**Random Forest**	**Random Forest**
Test Accuracy	**0.995**	**0.925**
Cross-Validation	**Up to 0.9988**	**0.847 (SVM CV)**
Training Accuracy	**~1.0**	**RF: 1.0 / SVM: 0.857**
Evaluation Reliability	**Potentially inflated (due to leakage)**	**More realistic and reliable**

Table 2: Comparison Between Kaggle Implementation and Our Study

The Kaggle implementation reports higher accuracy as it retains all available features, including those that may introduce target leakage. In contrast, this study removes leakage-prone features to ensure a fair and realistic evaluation of model performance. As a result, although the achieved accuracy is lower, the findings are more reliable and better reflect real-world predictive capability.

4.0 Description of the Proposed Techniques

4.1 Technique #1 Random Forest

4.1.1 What is Random Forest?

Random Forest is a supervised machine learning algorithm commonly used for classification and prediction tasks. It is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting [7]. The algorithm works by generating several decision trees during the training phase and selecting the final prediction based on majority voting.

Random Forest is widely used in predictive maintenance and industrial applications because of its ability to handle complex datasets, noisy data, and non-linear relationships between features [8].

4.1.2 How Random Forest was Used in the Project

In this project, the Random Forest algorithm was implemented to classify pipeline conditions based on engineered features extracted from the dataset. Several preprocessing and feature engineering techniques were applied before training the model.

The model performance was evaluated using accuracy score, classification report, and confusion matrix. The Random Forest model achieved an overall accuracy of 92.5%, demonstrating strong classification performance on the testing dataset.

Further technical details about Random Forest algorithms and implementation can be found in recent machine learning literature and the Scikit-learn documentation [9].

4.2 Technique #2 Support Vector Machine (SVM)

4.2.1 What is Support Vector Machine?

Support Vector Machine (SVM) is a widely utilized nonparametric classification method grounded in computational learning theory and the principle of structural risk minimization. At its fundamental level, the algorithm functions by identifying an optimal decision boundary, known as the maximum-margin hyperplane, which separates distinct data classes within a high-dimensional feature space. To handle complex, non-linear patterns that cannot be separated in the original input space, SVM employs kernel functions to transform the data into a higher-dimensional space where linear separation becomes feasible. This approach focuses on minimizing the upper bound of generalization error rather than just training error, ensuring that the resulting model is mathematically optimal and capable of performing reliably on new datasets [6].

4.2.2 How SVM used in the project

The SVM algorithm is implemented as a Support Vector Classifier (SVC) to predict pipeline failures. The project utilizes a linear kernel, which, as noted by, is used when the data characteristics allow for linear separation without the need for more complex non-linear mappings. Additionally, the implementation incorporates a balanced class weight and a regularization parameter (C=1) to manage the trade-off between maximizing the margin and minimizing misclassification, a central optimization challenge in SVM theory [6].

5.0 Empirical Studies

Empirical studies in predictive maintenance have shown that machine learning techniques play a significant role in identifying early signs of equipment degradation. Several research works demonstrated that models such as Random Forest, Support Vector Machine (SVM), and Neural Networks can effectively classify the health condition of industrial components by analyzing operational and degradation-related features. In the oil and gas sector, previous studies highlighted the importance of using sensor data—such as pressure, temperature, corrosion impact, and material loss—to detect pipeline failures before they occur. These empirical findings confirm that data-driven approaches can reduce unexpected breakdowns, improve safety, and optimize maintenance planning.
Based on these studies, this project applies machine learning models to classify pipeline segments into Normal, Moderate, and Critical states using real-world operational data.

5.1 Description of dataset

The dataset used in this project consists of 1000 pipeline records and includes 11 attributes that describe both the operational conditions and degradation characteristics of oil and gas pipelines. The dataset is structured and suitable for supervised multi-class classification, where the target variable is the pipeline Condition (Normal, Moderate, Critical).

The dataset contains numerical and categorical features that represent pipeline dimensions, material properties, operating environment, and degradation indicators. These features provide a comprehensive view of pipeline health and support the development of predictive maintenance models.

The attributes included in the dataset are described below:

Attribute	Description
Pipe_Size_mm	The diameter of the pipeline in millimeters
Thickness_mm	The wall thickness of the pipeline
Material	The material type used in the pipeline construction
Grade	The quality grade of the pipeline material
Max_Pressure_psi	The maximum pressure the pipeline can withstand (PSI)
Temperature_C	The operating temperature in degrees Celsius
Corrosion_Impact_Percent	The percentage of corrosion impact on the pipeline
Thickness_Loss_mm	The amount of wall thickness lost due to corrosion
Material_Loss_Percent	The percentage of material loss caused by degradation
Time_Years	The age of the pipeline in years
Condition (Target)	The health state of the pipeline: Normal, Moderate, or Critical

Table 3: description of data set attributes

5.1.1 Statistical Analysis of the Dataset

The statistical analysis of the dataset is presented in table 5 below. The mean, median, standard deviation, maximum and minimum values of the dataset are presented.

Attribute	Mean	Median	Standard Deviation	Maximum	Minimum
pipe_size_mm	522.00	300.00	443.70	1500.00	50.00
thickness_mm	16.07	12.93	10.55	49.53	3.00
max_pressure_psi	1004.10	900.00	812.84	2500.00	150.00
temperature_c	42.60	41.20	41.13	149.70	-50.00
corrosion_impact_percent	9.75	9.72	5.82	20.00	0.00
thickness_loss_mm	4.89	4.92	2.90	9.99	0.01
material_loss_percent	46.75	31.66	46.60	318.75	0.08
time_years	12.96	13.00	7.14	25.00	1.00

Table 4: Statistical Analysis of the Dataset

Attribute	Correlation with Condition
Thickness Loss (mm)	-0.9151
Material Loss (%)	-0.5292
Pipe Size (mm)	-0.0498
Thickness (mm)	-0.0338
Max Pressure (psi)	-0.0065
Corrosion Impact (%)	-0.0001
Temperature (°C)	0.0034
Time (years)	0.0016

Table 5: Correlation between each Attribute and the Encoded Target Variable

5.2 Experimental Setup

5.2.1 Dataset and Preprocessing

The dataset used in this project was obtained from Kaggle and contains 1,000 records related to oil and gas pipeline maintenance. It includes operational and physical pipeline features such as pipe size, thickness, material, grade, maximum pressure, temperature, corrosion impact percentage, thickness loss, material loss percentage, time in years, and pipeline condition.

Before training the machine learning models, the dataset was inspected to analyze its structure, data types, missing values, and duplicated records. The dataset contained no missing values and no duplicated rows, which minimized the need for extensive data cleaning. In addition, column names were standardized by converting them to lowercase and replacing spaces with underscores to simplify data processing in Python.

Categorical variables such as material, grade, and the target variable condition were transformed into numerical values using label encoding because machine learning models require numerical input data. Feature scaling was also applied using StandardScaler to normalize feature values before model training, especially since the Support Vector Machine model is sensitive to differences in feature scales.

In addition, leakage-related features were identified and removed before training to ensure realistic model evaluation and prevent unrealistically high prediction performance. This step helped improve the reliability and generalization capability of the models on unseen data.

5.2.2 Feature Engineering

Feature engineering was applied to improve the predictive ability of the models by creating new meaningful features from the original dataset. The goal was to provide the models with more informative indicators related to pipeline degradation and structural condition.

The engineered features included:

- **Thickness_Remaining_Pct:** represents the remaining wall thickness percentage after thickness loss.
- **Corrosion_Rate_Per_Year:** measures the annual corrosion rate based on thickness loss and time
- **Pressure_Per_Thickness:** represents the pressure applied relative to pipe thickness
- **Loss_Ratio:** shows the relationship between material loss and corrosion impact.
- **Pipe_Wall_Ratio:** represents the ratio between pipe thickness and pipe size.

These features were created because the EDA showed that individual variables such as pressure, temperature, and corrosion impact did not clearly separate the pipeline conditions on their own. Therefore, combining related variables helped the models capture more complex patterns in the data.

5.2.3 Train-Test Split and Cross-Validation

The dataset was divided into training and testing sets using an **80/20 split**, where 80% of the data was used for training and 20% was used for testing. The split was performed using stratify=y to preserve the original distribution of pipeline condition classes in both sets.

A fixed random_state = 42 was used to ensure reproducibility of the results.
In addition, **Stratified K-Fold Cross-Validation** with 5 splits was applied during training. This helped evaluate the models more reliably by ensuring that each fold maintained the same class distribution. Cross-validation was important because the dataset showed class imbalance, where some pipeline conditions appeared more frequently than others.

5.2.4 Model Configuration

1. Random Forest (RF)

The Random Forest model was implemented as the first machine learning technique for pipeline condition classification. It was selected because it can handle complex and non-linear relationships between features and is generally robust against noise and overfitting.

Hyperparameter tuning was performed by testing different values for n_estimators, min_samples_split, and min_samples_leaf. Based on the experiments, the final Random Forest configuration was:

- n_estimators = 200
- max_depth = None
- min_samples_split = 4
- min_samples_leaf = 1
- class_weight = 'balanced'
- random_state = 42
- n_jobs = -1

The class_weight = 'balanced' parameter was used to reduce the effect of class imbalance by giving more importance to minority classes. The model achieved a cross-validation accuracy of approximately **88.87%** and a testing accuracy of **92.5%**.

2. Support Vector Machine (SVM)

Linear kernel chosen after initial experiments showed non-linear kernels (RBF) did not improve accuracy and increased overfitting.

Hyperparameters:

- kernel = 'linear'

- C = 1.0 (regularization strength)
- class_weight = 'balanced'
- random_state = 42

5.2.5 Evaluation Metrics

The models were evaluated using several performance metrics to provide a complete understanding of their classification performance. The main evaluation metrics included accuracy, precision, recall, F1-score, and confusion matrix.

Accuracy was used to measure the overall percentage of correct predictions. Precision measured how many predicted samples for each class were actually correct, while recall measured how many actual samples from each class were correctly identified. The F1-score provided a balance between precision and recall. The confusion matrix was used to show the number of correct and incorrect predictions for each class.

For the Random Forest model, the testing accuracy reached **92.5%**, with strong precision, recall, and F1-score values across the three classes. The confusion matrix showed that most samples were correctly classified, with only a small number of misclassifications. In comparison, the SVM model achieved **82.5%** testing accuracy, which was lower than Random Forest.

5.2.6 Experimental Workflow

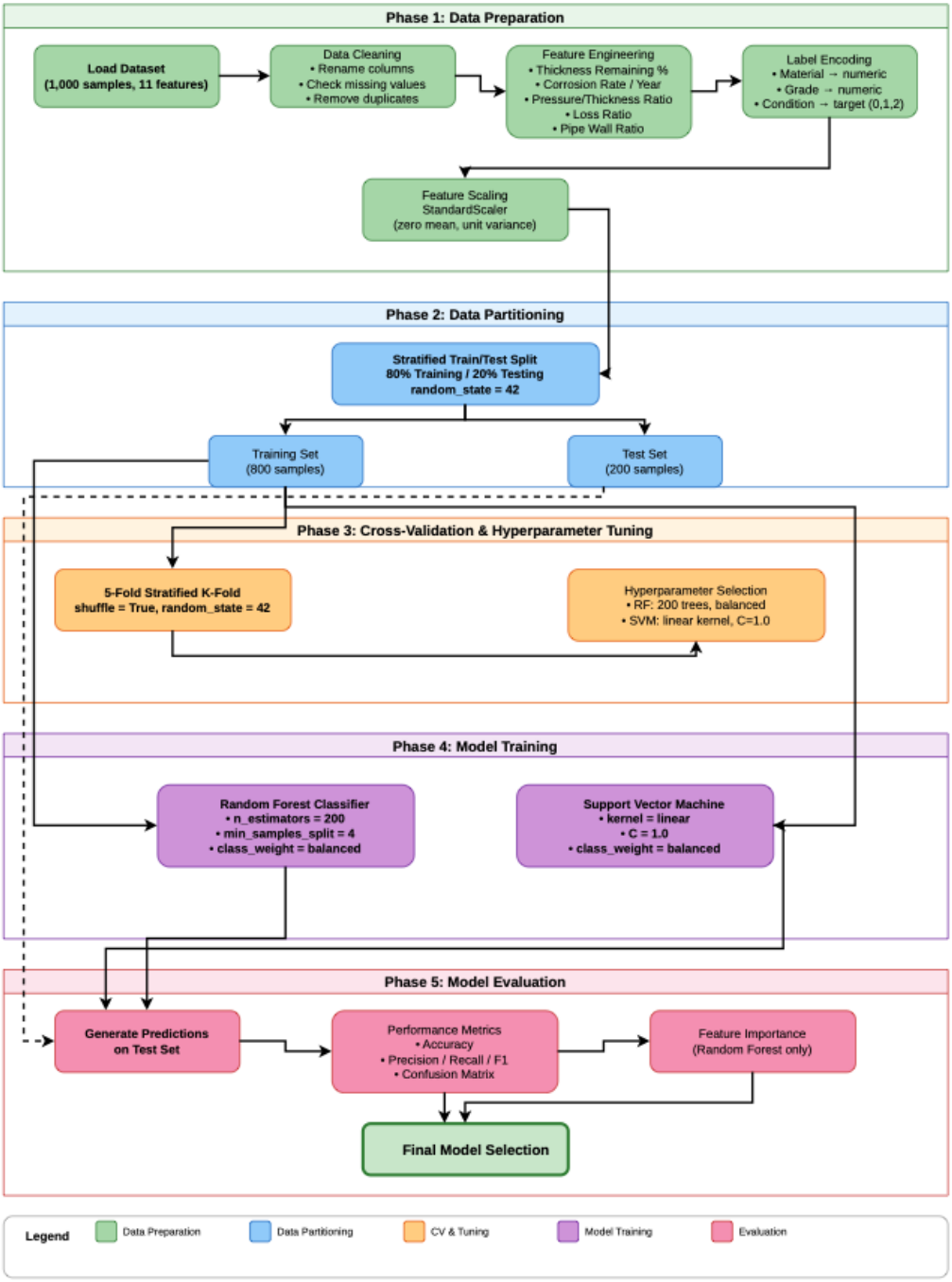

Figure 1: Experimental Workflow

5.3 Performance Measures

The performance evaluation showed that the Random Forest model outperformed the Support Vector Machine model for the oil and gas pipeline prediction dataset. The Random Forest model achieved a testing accuracy of 92.5%, while the SVM model achieved 82.5% accuracy. Therefore, Random Forest was selected as the final classification model for this dataset.

Random Forest was more suitable for this dataset because the dataset contains an imbalanced target class. Random Forest supports handling class imbalance through class weighting and bootstrap aggregation, making it more reliable for sensitive decision-making for applications such as oil and gas analysis where classification errors may lead to significant operational risks. In addition, Random Forest reduces overfitting by combining multiple decision trees and averaging their predictions, which improves generalization performance on smaller datasets.
Studies [10],[11] have also shown that Random Forest performs effectively on small and imbalanced datasets compared with many traditional classifiers.

Although SVM performs well in many classification tasks, its performance is highly dependent on selecting the correct kernel and hyperparameters. For imbalanced datasets, SVM may become biased toward the majority class which did happen in our case.[12]

5.3.1 Chosen Parameters for Models Against Accuracy Graphs

The following graphs illustrate the relationship between the selected hyperparameters and the obtained accuracy. For all graphs, the parameter with the highest accuracy was chosen.

Random Forest Model:

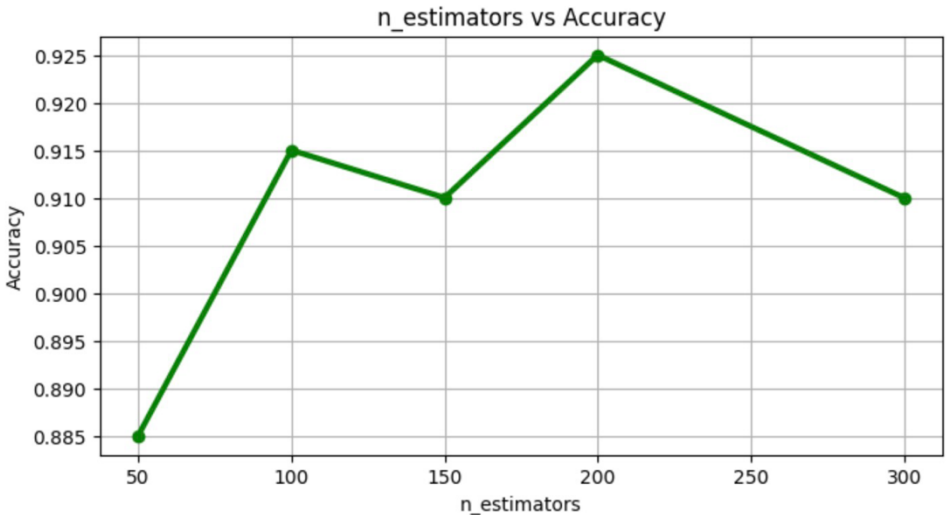

Figure 2: n_estimators vs Accuracy

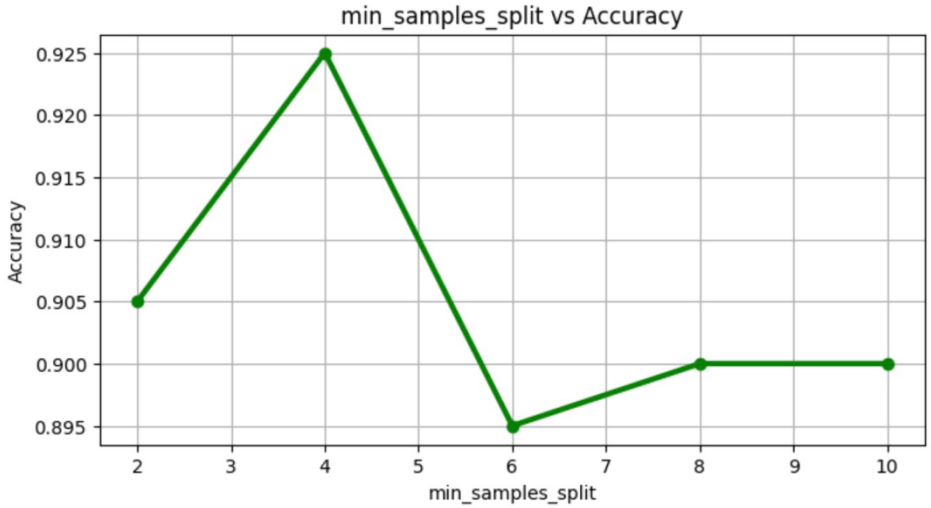

Figure 3: min_samples_split vs Accuracy

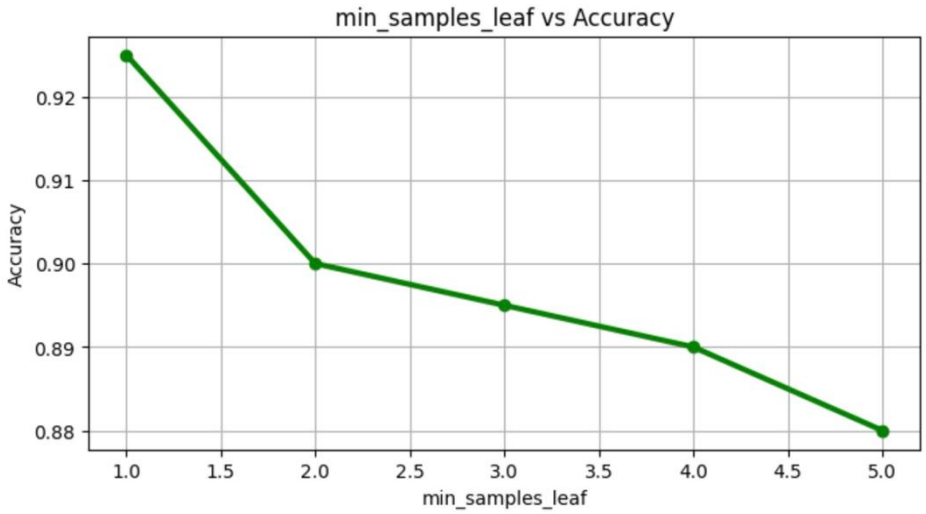

Figure 4: min_samples-_leaf vs Accuracy

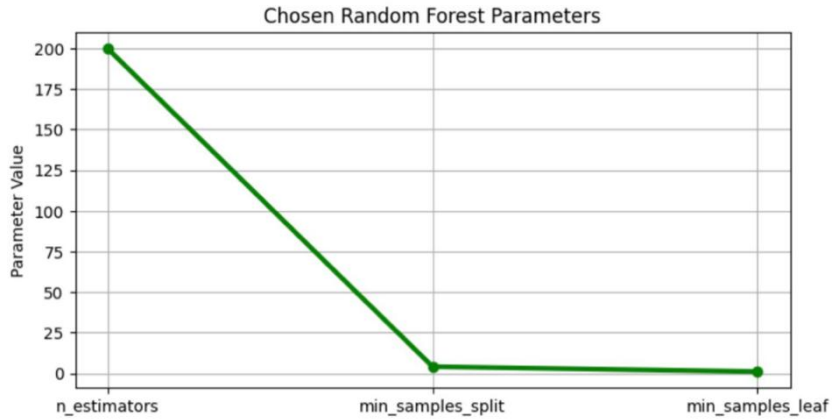

figure 5: random forest parameters

SVM Model:

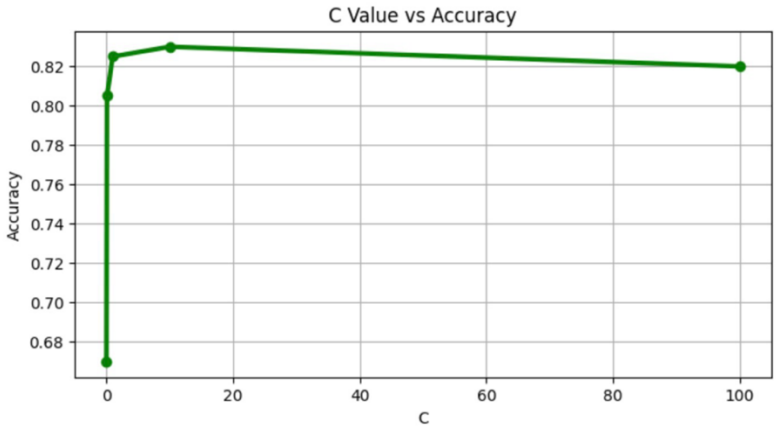

Figure 6: c vs Accuracy

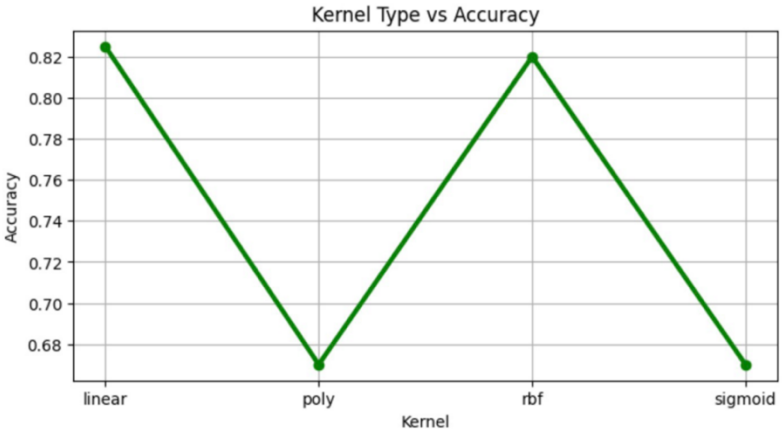

Figure 7: kernal type vs Accuracy

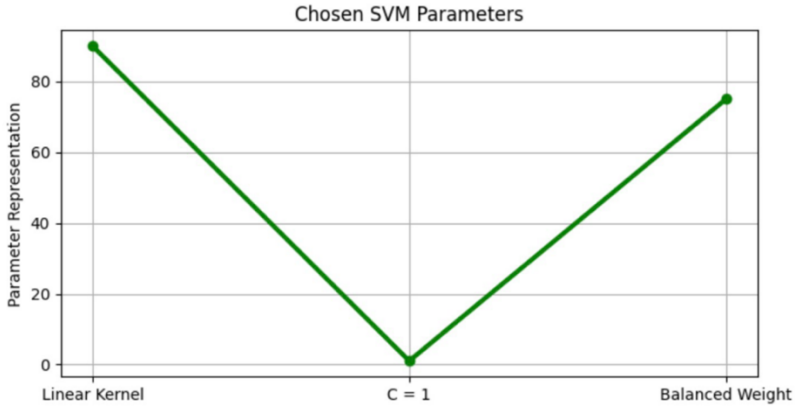

Figure 8: SVM parameters

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The implementation code for the machine learning project is available in the following GitHub repository:

https://github.com/Noor-Alshaib/Machine-learning-project-code.git